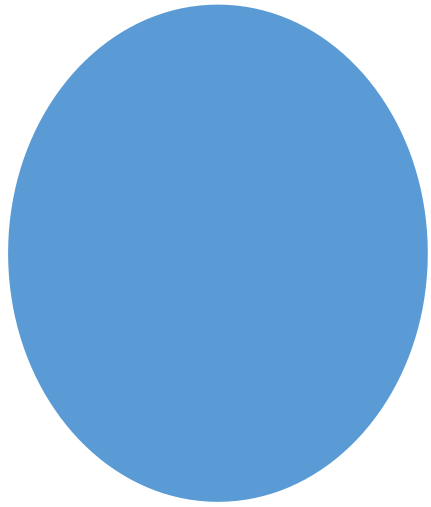


Data Visualization



- Descriptive Statistics
 - Simple visualization are useful. But what to use?
 - Min, max, mean, median, variance.
- Visualisation
 - Pie chart, bar chart, histogram, line graphs, etc.

Roadmap

- 'Raw' data hardly reveal any information.
- How should we 'make sense' from them?
- Descriptive statistics:
 - A suit of methods aim to reveal important characteristics of the data.
 - Two main approaches: numeric and visual descriptions!
- **An important consideration is the type of data!!!**
 - Step 1: Use data type classification tree.
 - Step 2: Apply appropriate descriptive methods.

Descriptive Statistics

Understand nominal data

- Numeric description:
 - Frequency
 - Normalized Frequency
 - Mode, ~~Mean~~, ~~Median~~

Frequency

Hair Color	Frequency
Brown	2
Red	3
Black	5



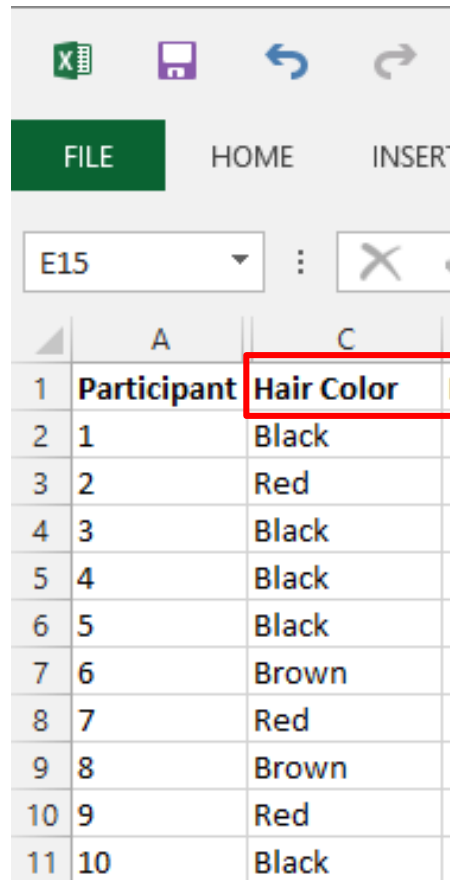
Participant	Hair Color
1	Black
2	Red
3	Black
4	Black
5	Black
6	Brown
7	Red
8	Brown
9	Red
10	Black

Nominal attribute

- Count the number of occurrences.
- Also known as countable.
- Order does not matter for nominal data, hence order of name appearance is not important.
- However, to improve comprehension we should sort the frequency!

Understand nominal data

Frequency



	A	C
1	Participant	Hair Color
2	1	Black
3	2	Red
4	3	Black
5	4	Black
6	5	Black
7	6	Brown
8	7	Red
9	8	Brown
10	9	Red
11	10	Black

Hair Color	Frequency
Brown	2
Red	3
Black	5

SORTING
the
FREQUENCY



Hair Color	Frequency
Black	5
Red	3
Brown	2

- Count the number of occurrences.
- Also known as counttable.
- Order does not matter for nominal data, hence order of name appearance is not important.
- However, to improve comprehension we should sort the frequency!

Understand nominal data

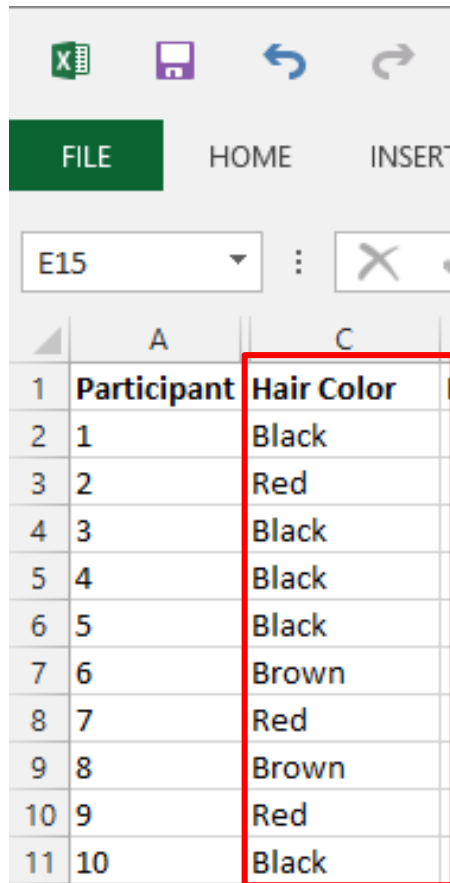
Hair Color	Frequency	Normalized Frequency (%)
Black	5	50
Red	3	30
Brown	2	20

$$\frac{5}{10} \times 100 = 50$$

□ Normalised Frequency

1. Count the number of occurrences.
2. Then normalised to the total!

- Also known as relative or percentage frequency table.

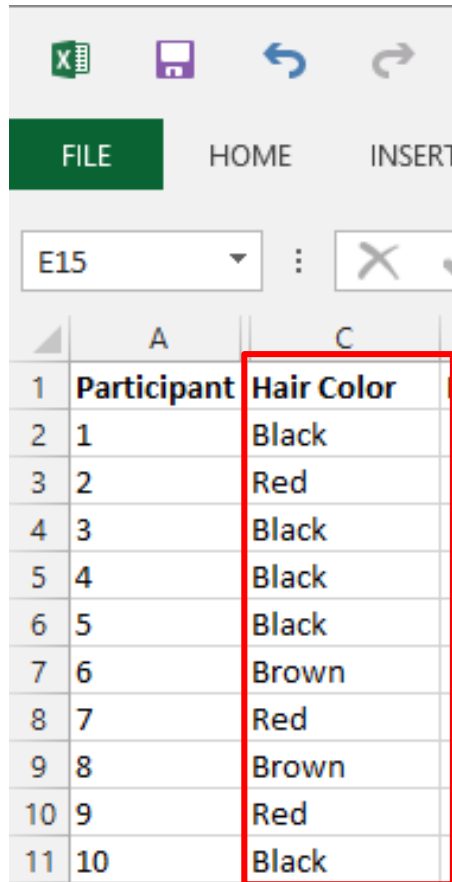


	A	C
1	Participant	Hair Color
2	1	Black
3	2	Red
4	3	Black
5	4	Black
6	5	Black
7	6	Brown
8	7	Red
9	8	Brown
10	9	Red
11	10	Black

What is the proportion of participants having black hair?

Understand nominal data

Hair Color	Frequency	Normalized Frequency (%)
Black	5	50
Red	3	30
Brown	2	20



Participant	Hair Color
1	Black
2	Red
3	Black
4	Black
5	Black
6	Brown
7	Red
8	Brown
9	Red
10	Black

What is the most popular hair color?

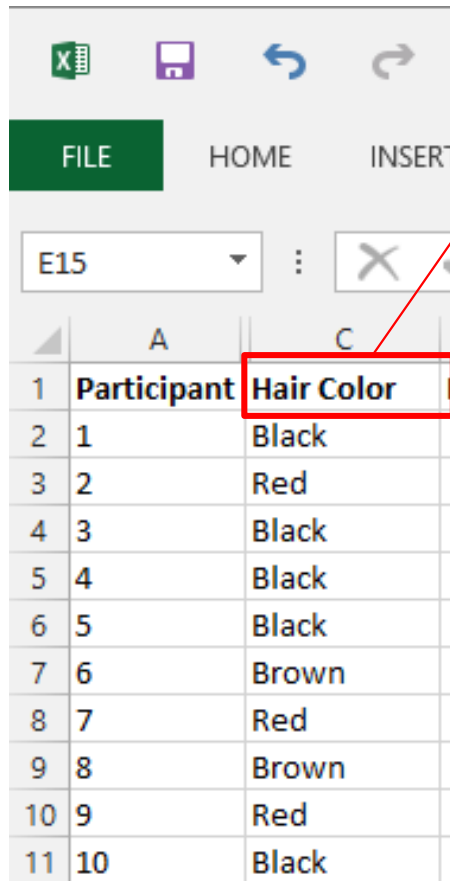
Answer: Black

Mode

- Label that has the highest frequency.
- What to do if we have more than one label attaining highest frequency?
 - Multiple modes
 - Report them all

Understand nominal data

Hair Color	Frequency	Normalized Frequency (%)
Black	5	50
Red	3	30
Brown	2	20



	A	C
1	Participant	Hair Color
2	1	Black
3	2	Red
4	3	Black
5	4	Black
6	5	Black
7	6	Brown
8	7	Red
9	8	Brown
10	9	Red
11	10	Black

Nominal attribute

- **Mode**
 - Label that has the highest frequency.
 - What to do if we have more than one label attaining highest frequency?
 - Multiple modes
 - Report them all
- Should not use: mean, median
- Why? because: there is no order!

Understand nominal data

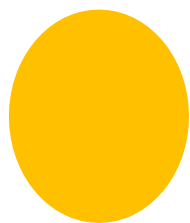
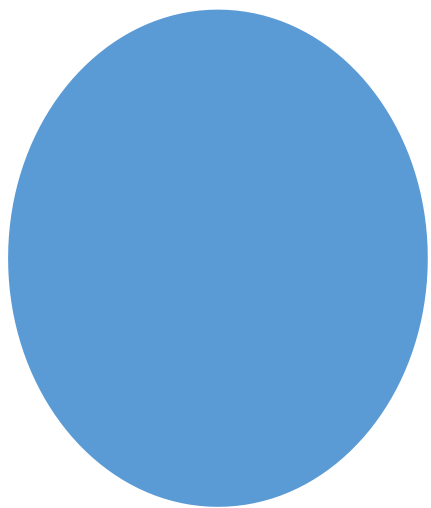
Understand nominal data

- Numeric description:

- Frequency
- Normalized Frequency
- Mode, Mean, Median

- Visual description:

- Pie chart (for small number of labels)
- TagCloud (for large number of labels)
- Bar Chart.
- Clustered/Stacked Bar Chart



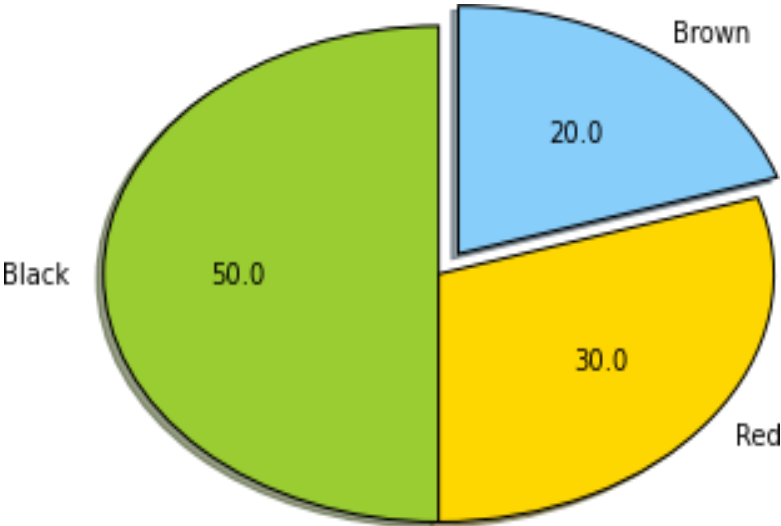
PIE CHART



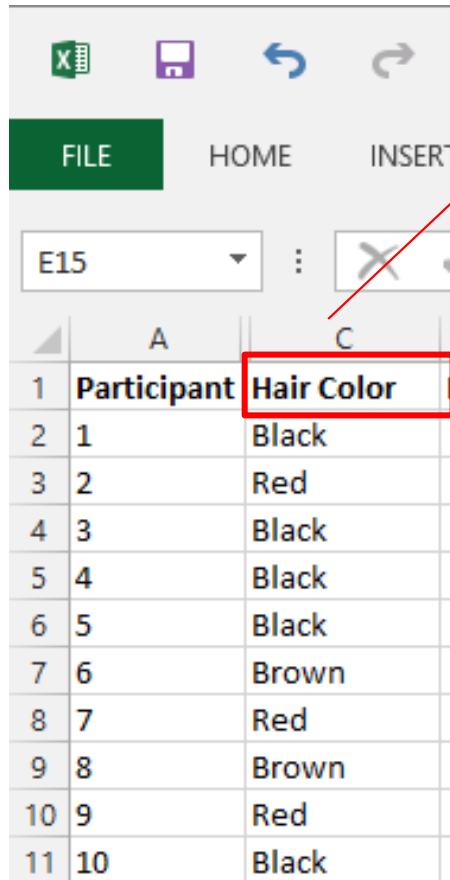
□ **Pie Chart**

- Divided into sectors.
- Describes the percentage for each label (out of the total).

Hair Color	Frequency	Normalized Frequency (%)
Black	5	50
Red	3	30
Brown	2	20



□ Pie Chart



	A	C
1	Participant	Hair Color
2	1	Black
3	2	Red
4	3	Black
5	4	Black
6	5	Black
7	6	Brown
8	7	Red
9	8	Brown
10	9	Red
11	10	Black

Nominal attribute

Hair Color	Frequency	Normalized Frequency
Black	5	50%
Red	3	30%
Brown	2	20%

```
labels = 'Black', 'Red', 'Brown'

# frequency count
xx = [5, 3, 2]

# colors
colors = ['yellowgreen', 'gold', 'lightskyblue']

# explode the third one
explode = (0, 0, 0.1)

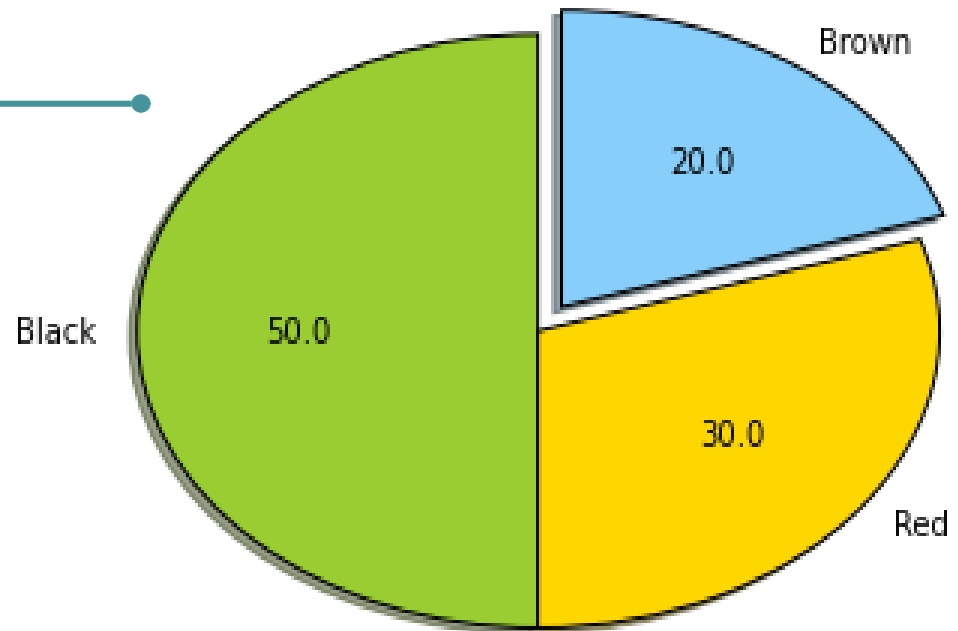
dummy=pie(xx,labels=labels,explode=explode, colors=colors,
          autopct='%1.1f', shadow=True, startangle=90)
```

□ Pie Chart

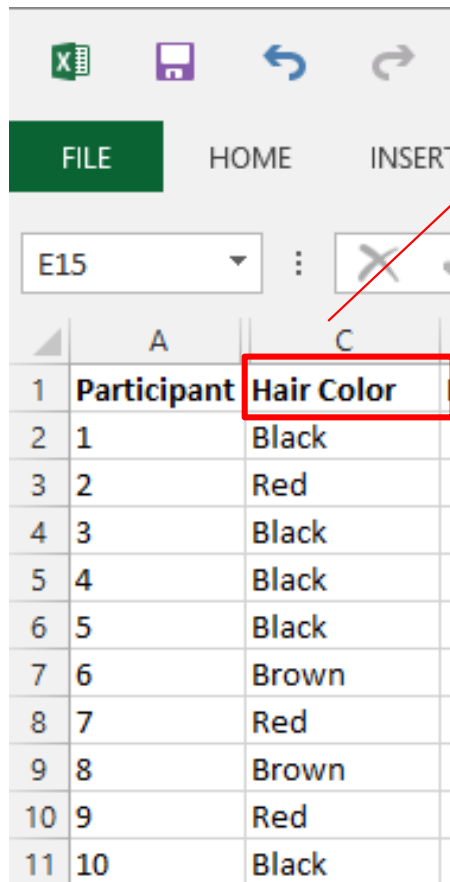
- Divided into sectors.
- Describes the percentage for each label (out of the total).

Hair Color	Frequency	Normalized Frequency (%)
Black	5	50
Red	3	30
Brown	2	20

It is correct, but can we still improve this plot? Visually?



□ Pie Chart



	A	C
1	Participant	Hair Color
2	1	Black
3	2	Red
4	3	Black
5	4	Black
6	5	Black
7	6	Brown
8	7	Red
9	8	Brown
10	9	Red
11	10	Black

Nominal attribute

Hair Color	Frequency	Normalized Frequency
Black	5	50%
Red	3	30%
Brown	2	20%

```
labels = 'Black', 'Red', 'Brown'

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          autopct='%1.1f', shadow=True, startangle=90)
```

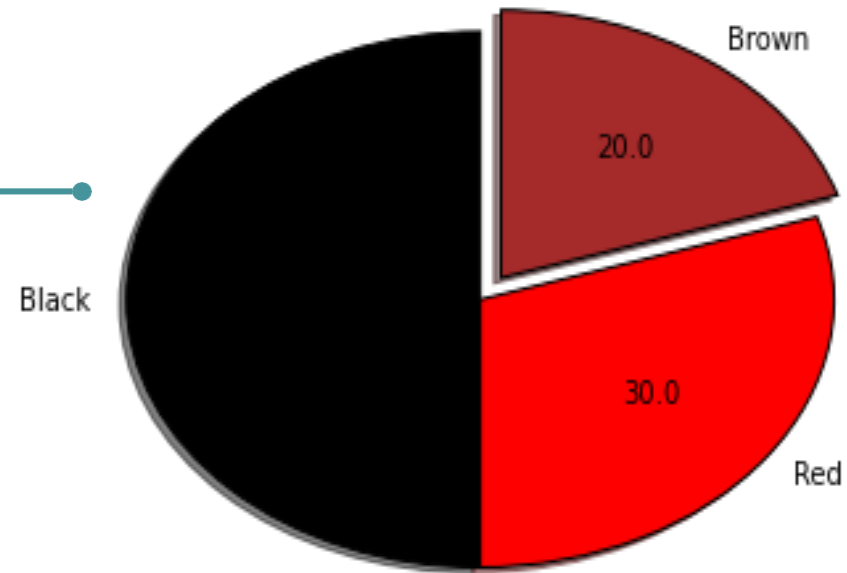
'black', 'red', 'brown'

□ Pie Chart

- Divided into sectors.
- Describes the percentage for each label (out of the total).

Hair Color	Frequency	Normalized Frequency (%)
Black	5	50
Red	3	30
Brown	2	20

It is correct, but can we still improve this plot? Visually?



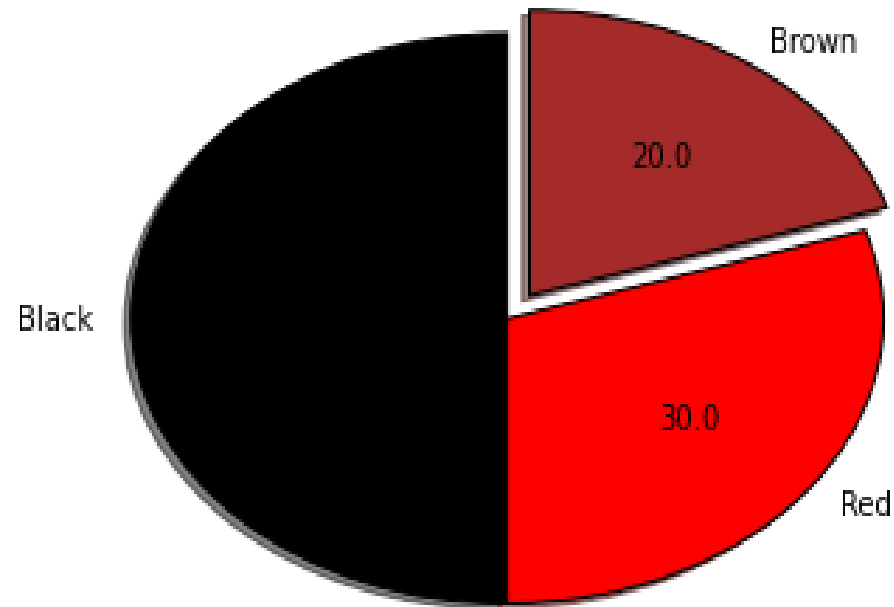
□ Pie Chart

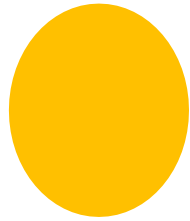
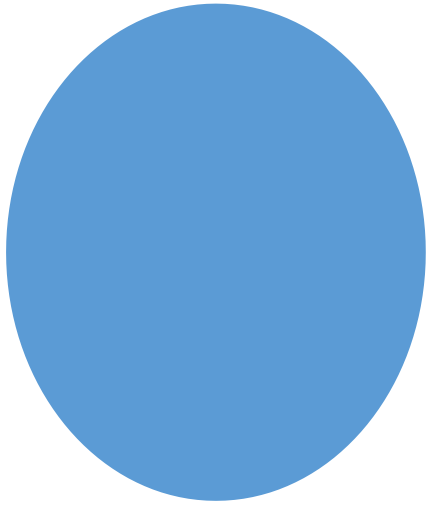
- Divided into sectors
- Each describes the percentage of the total

□ Properties

- Visually appeal.
- Gain understanding in distribution.
- But, only suitable for a small number labels, otherwise it will be very cluttered.

Hair Color	Frequency	Normalized Frequency (%)
Black	5	50
Red	3	30
Brown	2	20





TAG CLOUD



Field of Research	Volume	Percentage
HEALTH	2625.18	20.61%
ENGINEERING	1306.26	10.25%
COMPUTER SCIENCES	1187.04	9.32%
HUMAN SOCIETY	1166.04	9.15%
TOURISM & SERVICES	980.86	7.70%
EDUCATION	810.6	6.36%
CHEMISTRY	725.64	5.70%
BIOLOGY	678.79	5.33%
PSYCHOLOGY	436.6	3.43%
CREATIVE ARTS	404.33	3.17%
LINGUISTICS	348.01	2.73%
BUILT ENVIRONMENT	304.33	2.39%
HISTORY	294.19	2.31%
ECONOMICS	293.02	2.30%
PHILOSOPHY	282.31	2.22%
AGRICULTURE	228.21	1.79%
ENVIRONMENT	197.34	1.55%
TECHNOLOGY	164.06	1.29%
LAW	157	1.23%
MATHS	50.5	0.40%
EARTH SCIENCES	49.61	0.39%
PHYSICS	48.08	0.38%

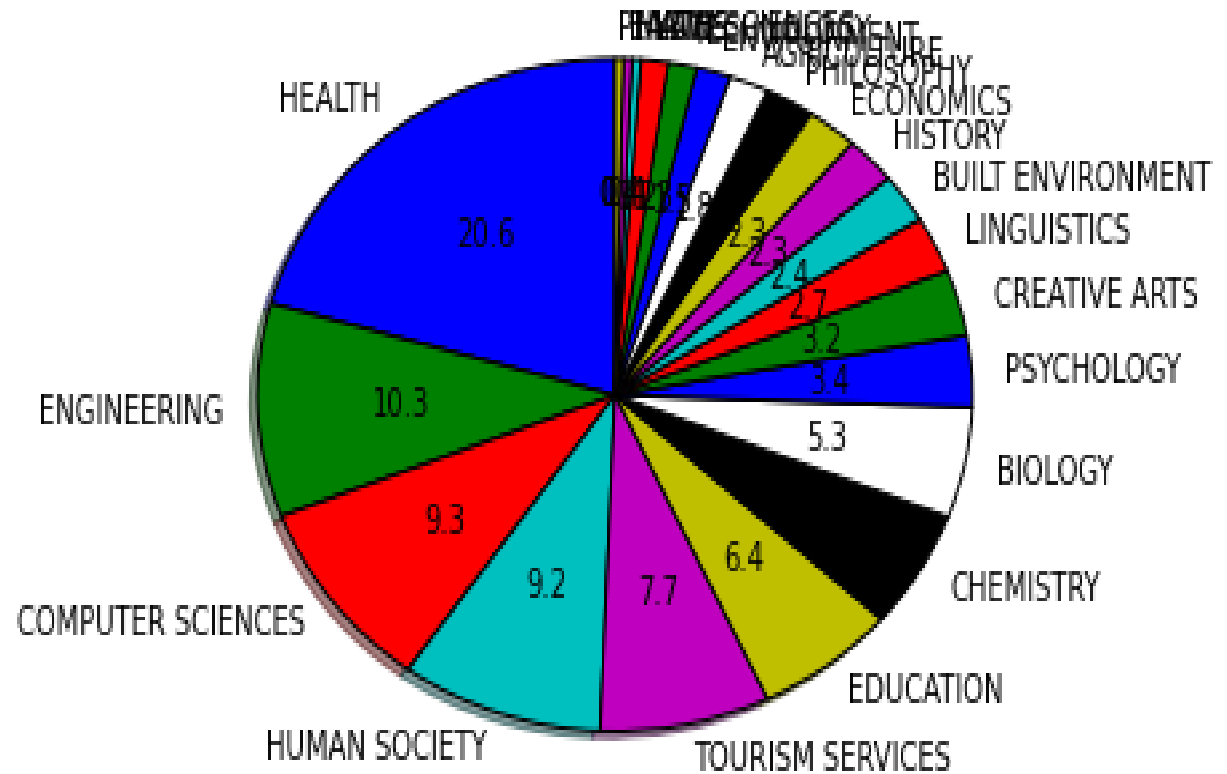
Excellence in Research

Australia (ERA) Data

- 22 Field of Research
- Calculate the volume

When Pie Chart fails?

Field of Research	Volume	Percentage
HEALTH	2625.18	20.61%
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MATHS	50.5	0.40%
EARTH SCIENCES	49.61	0.39%
PHYSICS	48.08	0.38%



- Hard to read.
- Visually ugly.
- Imagine what it looks like if we have hundreds of labels!!!

Tag Cloud

Field of Research	Volume	Percentage
HEALTH	2625.18	20.61%
ENGINEERING	1306.26	10.25%
COMPUTERSCIENCES	1187.04	9.32%
HUMAN SOCIETY	1166.04	9.15%
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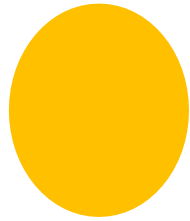
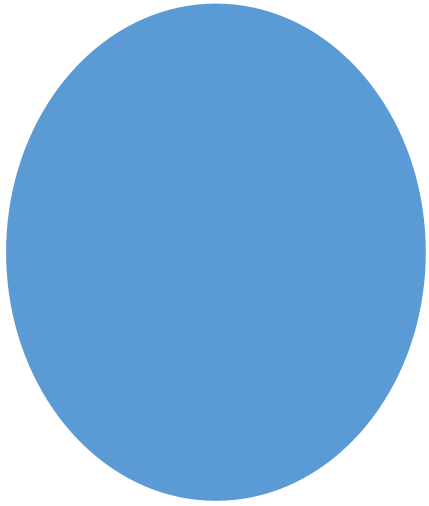


<http://wordle.net>



- The size of the label is proportional to the volume.
- Visually appealing.
- Limitation: we don't know precisely the proportion.

Tag Cloud



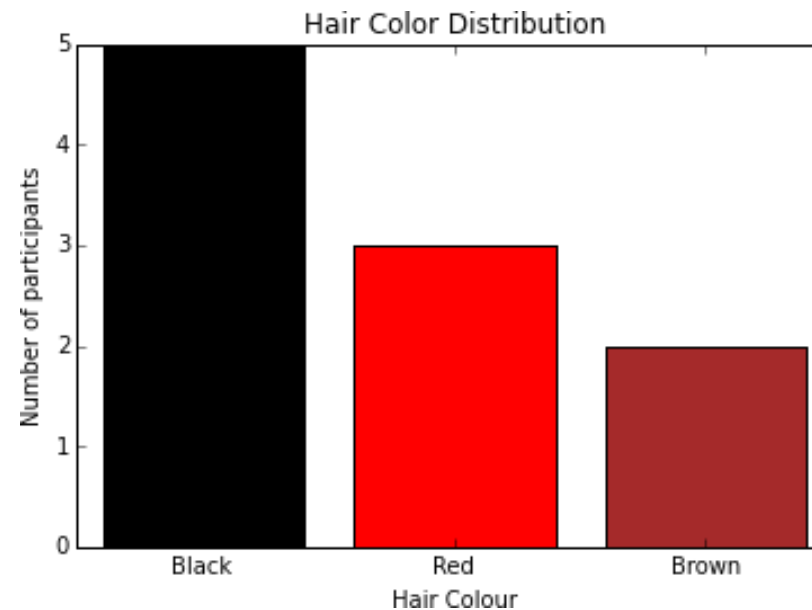
Bar Chart



- Another effective tool to visualise nominal data is the bar chart and its variants.
- Presentation:
 - Labels on the vertical axis.
 - Count/frequency/normalised frequency on the horizontal axis.
- There are three common types of bar charts:
 - **Simple bar chart**: to describe a single (nominal) variable.
 - **Clustered bar chart**: to describe two or more variables.
 - **Stacked bar chart**: to describe two or more variables.

Bar Chart

Participant	Gender	Hair Color
1	Male	Black
2	Female	Red
3	Male	Black
4	Male	Black
5	Male	Black
6	Female	Brown
7	Male	Red
8	Female	Brown
9	Female	Red
10	Female	Black



- Bars should have the same width.
- There are equal spaces between bars
 - To emphasise the categorical nature of the data.

Simple Bar Chart

```
import matplotlib.pyplot as plt

# plot a simple bar chart
hair = 'Black', 'Red', 'Brown'

# frequency count
freq = [5, 3, 2]
import numpy as np
import matplotlib.pyplot as plt

colors = ['black','red','brown']

x_pos = np.arange(len(hair))
plt.bar(x_pos, freq,align='center',color=colors)

plt.xticks(x_pos, hair)
plt.ylabel('Number of participants')
plt.xlabel('Hair Colour')
plt.title('Hair Color Distribution')
```

Participant	Gender	Hair Color
1	Male	Black
2	Female	Red
3	Male	Black
4	Male	Black
5	Male	Black
6	Female	Brown
7	Male	Red
8	Female	Brown
9	Female	Red
10	Female	Black

Clustered BarChart

- Visualise the statistics of one variable in relation to another variable.
- E.g., what is the distribution of hair colour among female?
- E.g., what is the distribution of gender for those having black hair?

contingency table

(aka., co-occurrence matrix, cross-tabular table, etc.)

Hair Color

	Gender	
	Male	Female
Black	4	1
Red	1	2
Brown	0	2

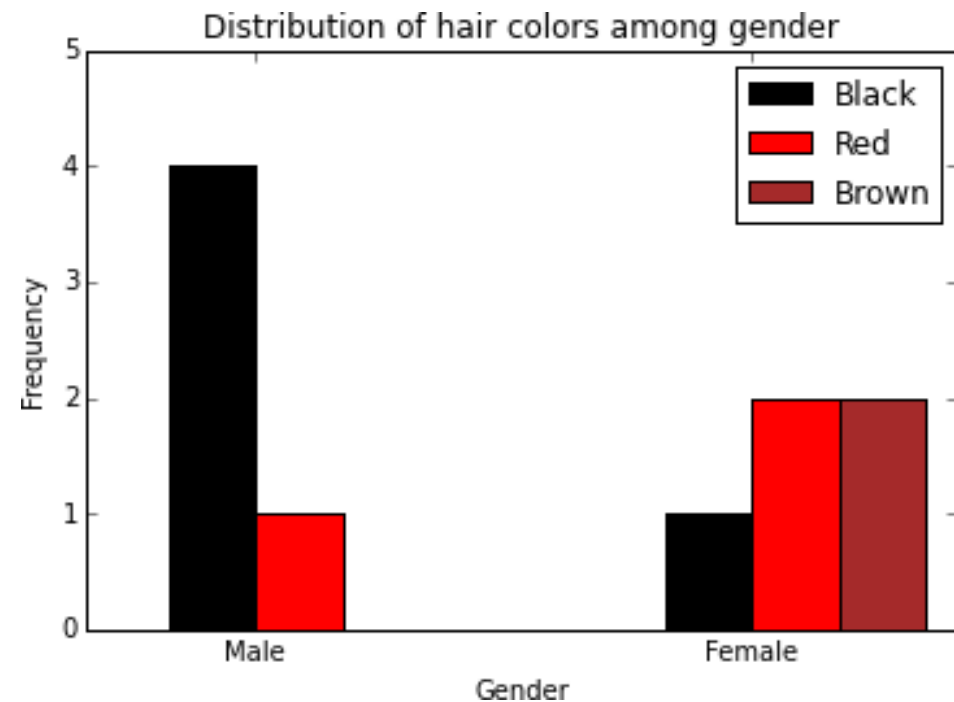
Clustered Bar Chart

Participant	Gender	Hair Color
1	Male	Black
2	Female	Red
3	Male	Black
4	Male	Black
5	Male	Black
6	Female	Brown
7	Male	Red
8	Female	Brown
9	Female	Red
10	Female	Black

Clustered BarChart

- Visualise the statistics of one variable in relation to another variable.
- E.g., what is the distribution of hair colour among female?

		Gender	
		Male	Female
Hair Color	Black	4	1
	Red	1	2
	Brown	0	2



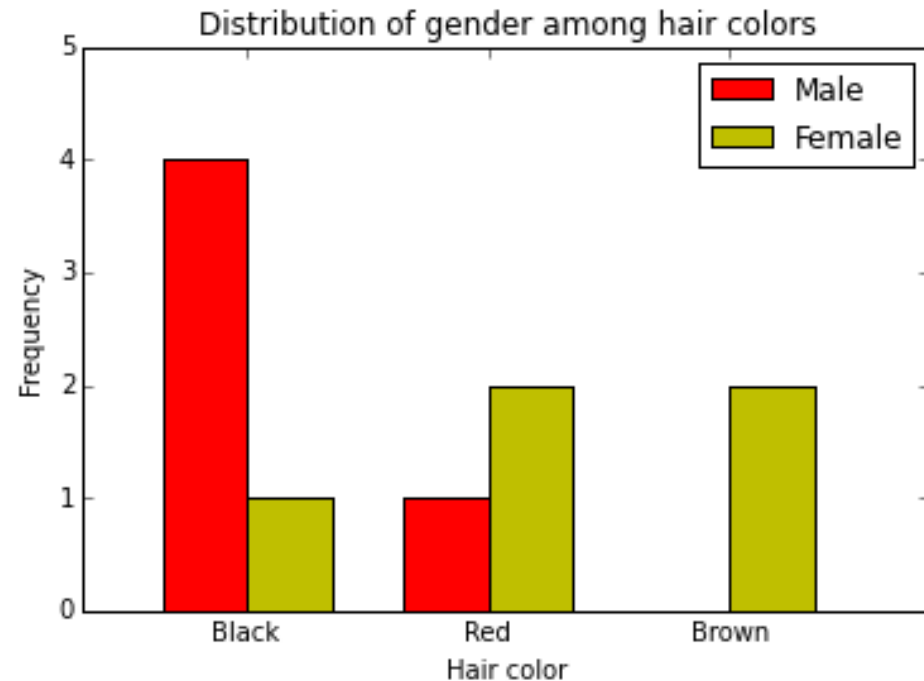
Clustered Bar Chart

Participant	Gender	Hair Color
1	Male	Black
2	Female	Red
3	Male	Black
4	Male	Black
5	Male	Black
6	Female	Brown
7	Male	Red
8	Female	Brown
9	Female	Red
10	Female	Black

Clustered BarChart

- Visualise the statistics of one variable in relation to another variable.
- E.g., what is the distribution of hair colour among female?
- E.g., what is the distribution of gender for those having black hair?

		Gender	
		Male	Female
Hair Color	Black	4	1
	Red	1	2
	Brown	0	2



Clustered Bar Chart

Participant	Gender	Hair Color
1	Male	Black
2	Female	Red
3	Male	Black
4	Male	Black
5	Male	Black
6	Female	Brown
7	Male	Red
8	Female	Brown
9	Female	Red
10	Female	Black

Gender

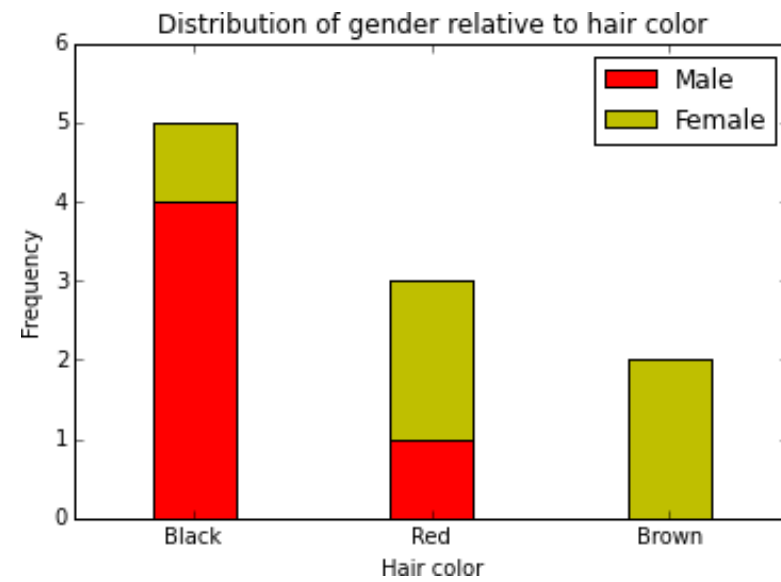
		Male	Female
Hair Color	Black	4	1
	Red	1	2
	Brown	0	2

Stacked Bar Chart

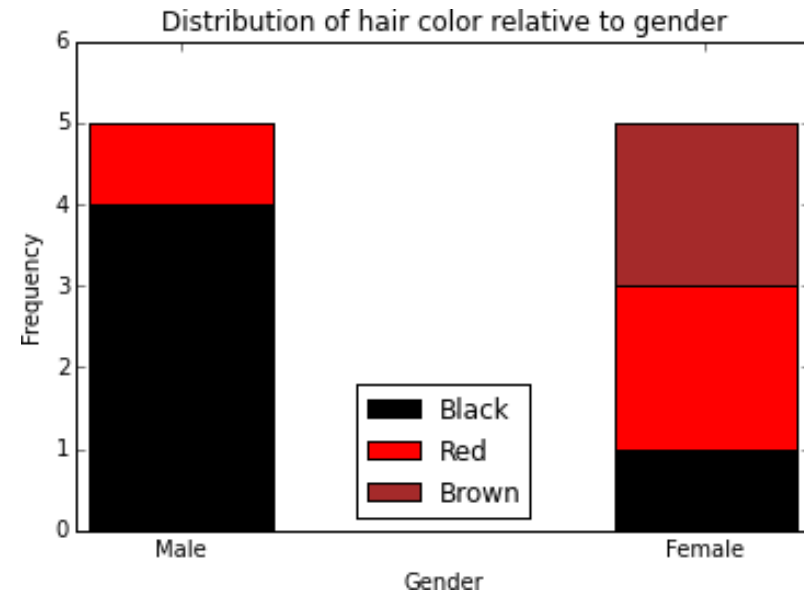
- Very similar to clustered bar chart.
- Visualise the statistics of one variable in relation to another variable.
- E.g., what is the distribution of hair colour among female?
- E.g., what is the distribution of gender for those having black hair?

Stacked Bar Chart

Participant	Gender	Hair Color
1	Male	Black
2	Female	Red
3	Male	Black
4	Male	Black
5	Male	Black
6	Female	Brown
7	Male	Red
8	Female	Brown
9	Female	Red
10	Female	Black



		Gender	
		Male	Female
Hair Color	Black	4	1
	Red	1	2
	Brown	0	2



Stacked Bar Chart

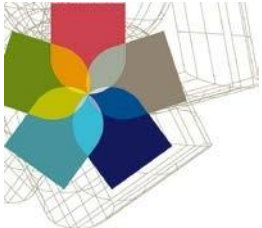
Summary for nominal data

□ Numeric description:

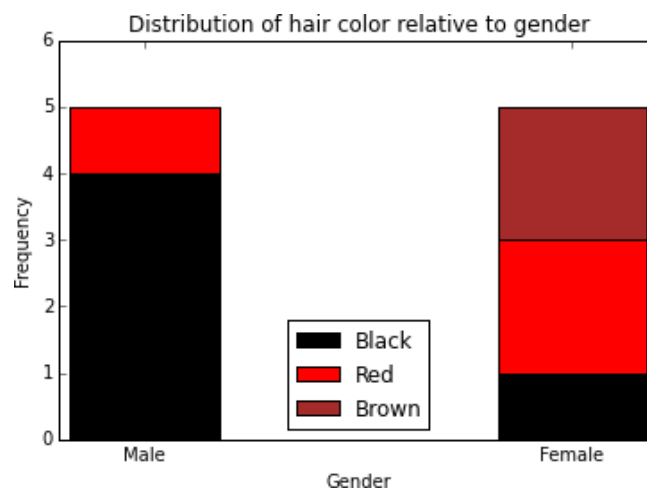
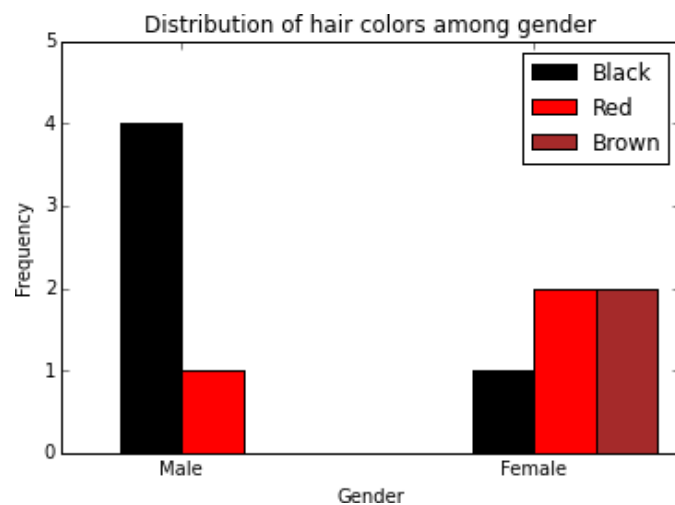
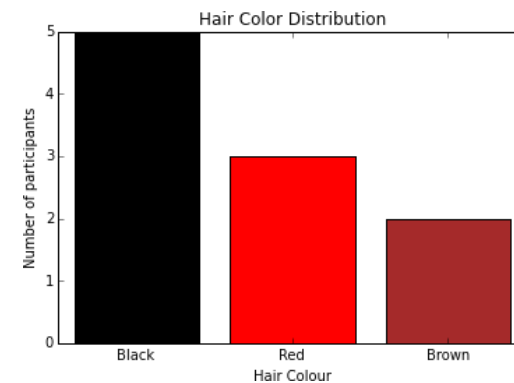
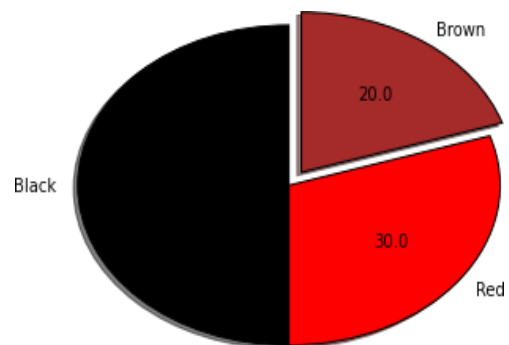
- Frequency
- Normalized Frequency
- Mode, ~~Mean~~, Median

□ Visual description:

- Pie chart (for small number of labels)
- TagCloud (for large number of labels)
- Bar Chart.
- Clustered/Stacked Bar Chart



Summary for nominal data



Understand Ordinal Data

□ Numeric description:

- Frequency
- Normalised Frequency
- Cumulative Frequency
- Cumulative Normalised Frequency.
- Mode, ~~Mean~~, Median

□ Visual description:

- Pie Chart
- Tag Cloud
- Bar Chart
- Clustered/Stacked Bar Chart
- Step Chart

Sex	Seatbelt
Male	Rarely
Female	Mosttimes
Female	Always
Male	Always
Female	Always
Female	Always
Female	Sometimes
Female	Always
Female	Mosttimes
Female	Rarely
Female	Mosttimes
Female	Always
Female	Mosttimes
Male	Always
Male	Mosttimes

Values must be sorted or ordered!

- Cumulative frequency:
 - Accumulated count.
 - Values must be ordered.
 - E.g., how many students who did **not** always wear seatbelt?
- Why it doesn't make sense to compute cumulative frequency for nominal data?

Seatbelt	Frequency	Cumulative Frequency
Never	0	0
Rarely	2	2=2+0
Sometimes	1	3=1+2
Mosttimes	5	8=5+3
Always	7	15=7+8

How many students who did **not** always wear seatbelt?

Answer: **8**

Cumulative Frequency

Sex	Seatbelt
Male	Rarely
Female	Mosttimes
Female	Always
Male	Always
Female	Always
Female	Always
Female	Sometimes
Female	Always
Female	Mosttimes
Female	Rarely
Female	Mosttimes
Female	Always
Female	Mosttimes
Male	Always
Male	Mosttimes

- Cumulative frequency:
 - Accumulated count.
 - E.g., how many students who did **not** always wear seatbelt?
- Why it doesn't make sense to compute cumulative frequency for nominal data?

Seatbelt	Frequency	Cumulative Frequency
Never	0	0
Rarely	2	$2 = 2 + 0$
Sometimes	1	$3 = 1 + 2$
Mosttimes	5	$8 = 5 + 3$
Always	7	$15 = 7 + 8$

How many students who had never or rarely worn seatbelt?

Answer: ???

Cumulative Frequency

Sex	Seatbelt
Male	Rarely
Female	Mosttimes
Female	Always
Male	Always
Female	Always
Female	Always
Female	Sometimes
Female	Always
Female	Mosttimes
Female	Rarely
Female	Mosttimes
Female	Always
Female	Mosttimes
Male	Always
Male	Mosttimes

□ Cumulative normalised frequency:

- Accumulated normalised frequency.
- E.g., What is the **proportion** of students who did **not** always wear seatbelt?

Seatbelt	Frequency	Normalised Frequency (%)	Cumulative Normalised Frequency (%)
Never	0	0=0/15	0
Rarely	2	13.3=2/15	13.3=13.3+0
Sometimes	1	6.7=1/15	20.0=6.7+13.3
Mosttimes	5	33.3=5/15	53.3=33.3+20.0
Always	7	46.7=7/15	100=46.7+53.3

What is the proportion of student who did not always wear seatbelt?

Answer: **53.3**
approximately a half

Cumulative Normalised Frequency

Sex	Seatbelt
Male	Rarely
Female	Mosttimes
Female	Always
Male	Always
Female	Always
Female	Always
Female	Sometimes
Female	Always
Female	Mosttimes
Female	Rarely
Female	Mosttimes
Female	Always
Female	Mosttimes
Male	Always
Male	Mosttimes

□ Cumulative normalised frequency:

- Accumulated normalised frequency.
- E.g., What is the **proportion** of students who did **not** always wear seatbelt?

Seatbelt	Frequency	Normalised Frequency (%)	Cumulative Normalised Frequency (%)
Never	0	0=0/15	0
Rarely	2	13.3=2/15	13.3=13.3+0
Sometimes	1	6.7=1/15	20.0=6.7+13.3
Mostimes	5	33.3=5/15	53.3=33.3+20.0
Always	7	46.7=7/15	100=46.7+53.3

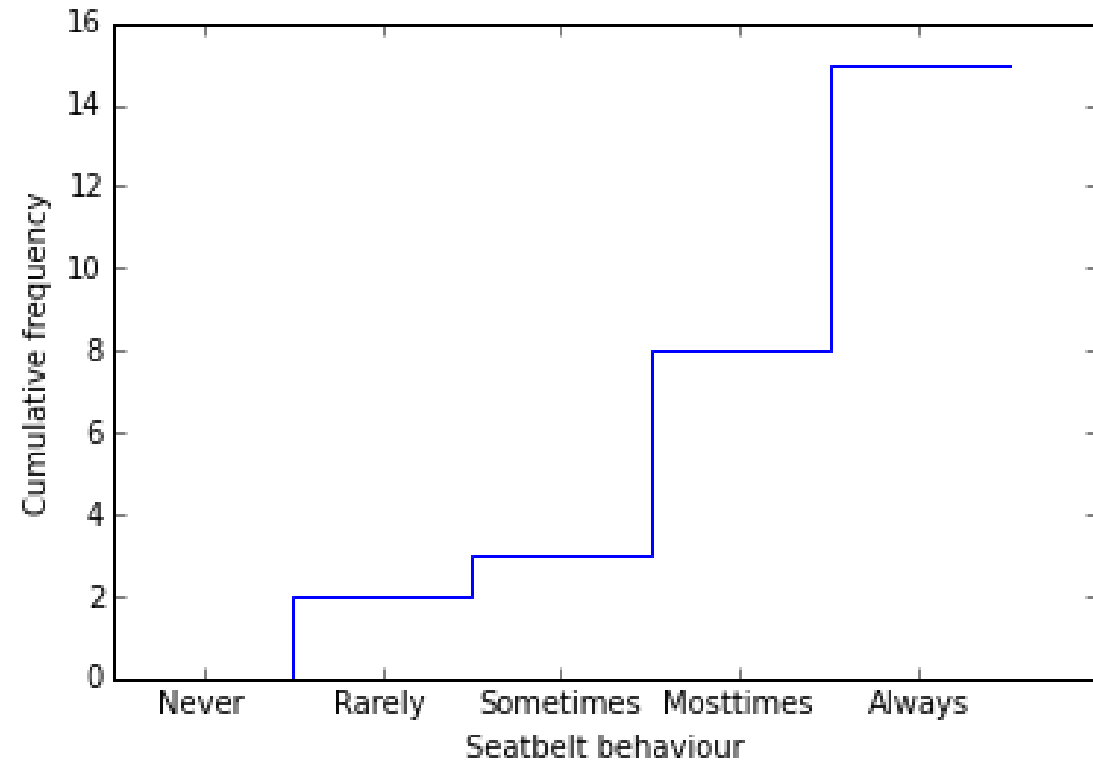
Why this number must always be added up to 100?

Cumulative Normalised Frequency

Sex	Seatbelt
Male	Rarely
Female	Mosttimes
Female	Always
Male	Always
Female	Always
Female	Always
Female	Sometimes
Female	Always
Female	Mosttimes
Female	Rarely
Female	Mosttimes
Female	Always
Female	Mosttimes
Male	Always
Male	Mosttimes

- Howto visualise cumulative quantities?
- Step chart:
 - Tovisualise a cumulative frequency or cumulative normalised frequency.

Seatbelt	Freque nc y	Cumulativ e Frequency
Never	0	0
Rarely	2	2
Sometimes	1	3
Mosttimes	5	8
Always	7	15



Step Chart

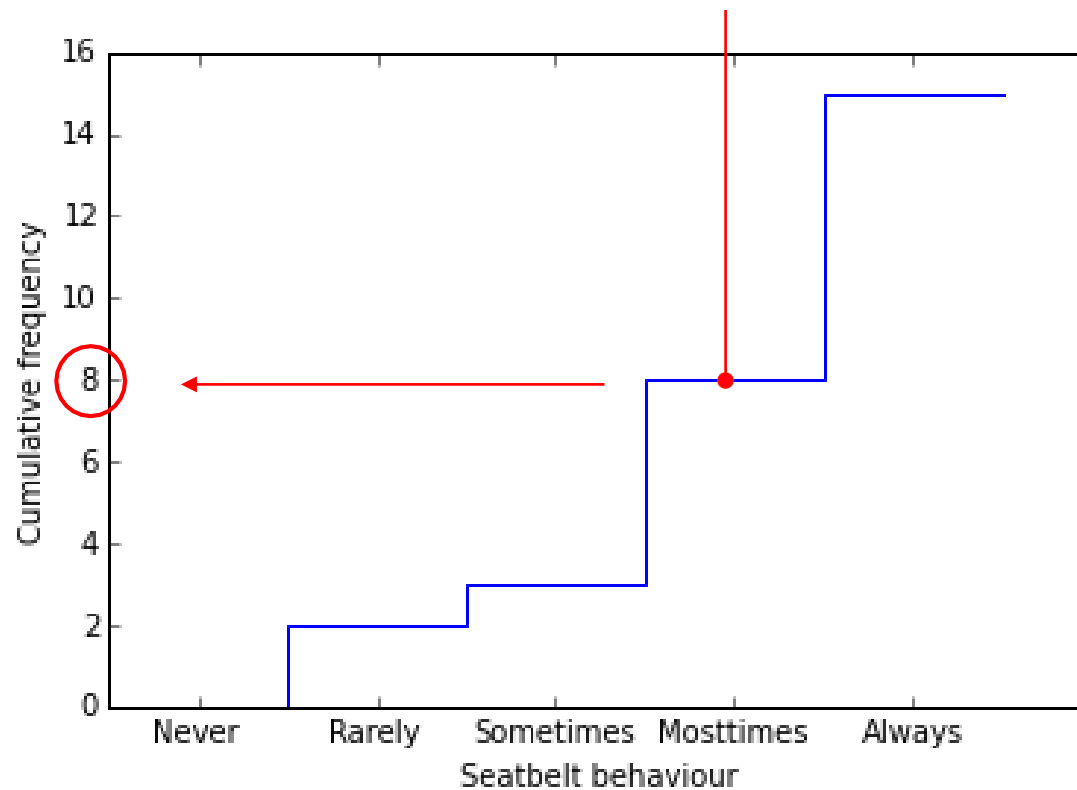
Sex	Seatbelt
Male	Rarely
Female	Mosttimes
Female	Always
Male	Always
Female	Always
Female	Always
Female	Sometimes
Female	Always
Female	Mosttimes
Female	Rarely
Female	Mosttimes
Female	Always
Female	Mosttimes
Male	Always
Male	Mosttimes

Answer: **8**

Step chart

- To visualise a cumulative frequency or cumulative normalised frequency.

How many students who did **not** always wear seatbelt?

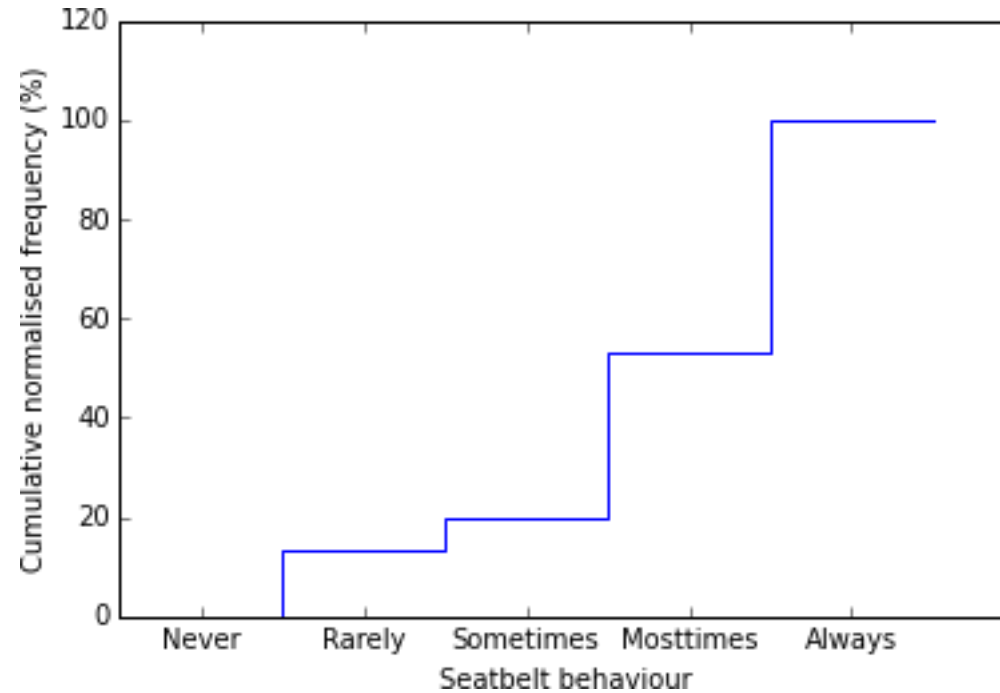


Step Chart

Sex	Seatbelt
Male	Rarely
Female	Mosttimes
Female	Always
Male	Always
Female	Always
Female	Always
Female	Sometimes
Female	Always
Female	Mosttimes
Female	Rarely
Female	Mosttimes
Female	Always
Female	Mosttimes
Male	Always
Male	Mosttimes

- Howto visualise cumulative quantities?
- Step chart:
 - Tovisualise a cumulative frequency or cumulative normalised frequency.

Seatbelt	Frequency	Cumulative Normalised Frequency (%)
Never	0	0
Rarely	2	13.3
Sometimes	1	20.0
Mosttimes	5	53.3
Always	7	100



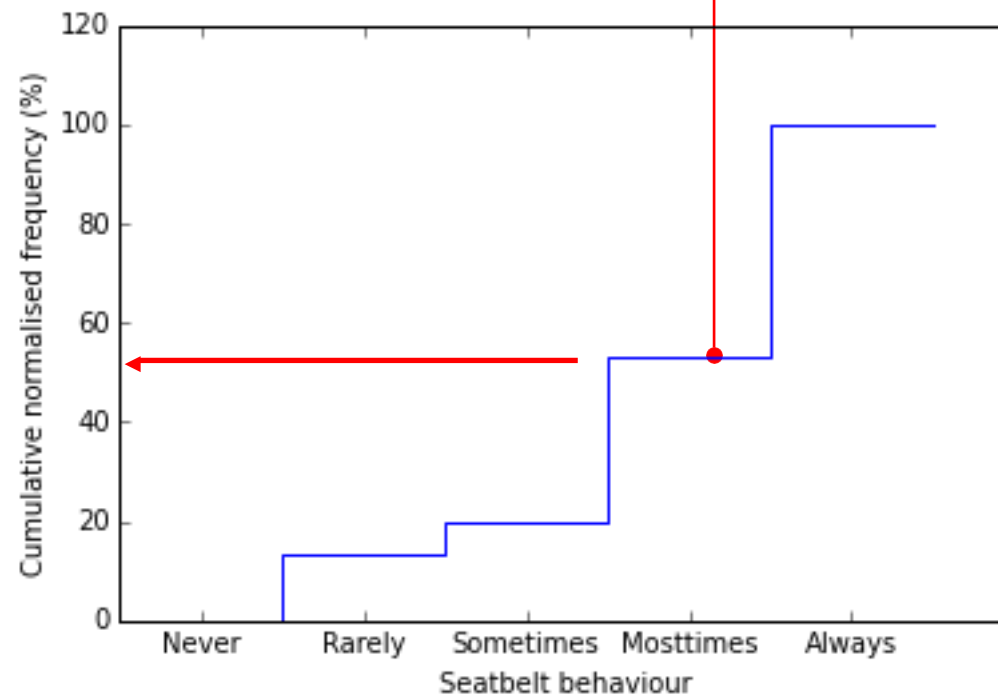
Step Chart

Sex	Seatbelt
Male	Rarely
Female	Mosttimes
Female	Always
Male	Always
Female	Always
Female	Always
Female	Sometimes
Female	Always
Female	Mosttimes
Female	Rarely
Female	Mosttimes
Female	Always
Female	Mosttimes
Male	Always
Male	Mosttimes

Step chart

- To visualise a cumulative frequency or **cumulative normalised frequency**.

What is the proportion of student who did not always wear seatbelt?



Answer: **53.3**
approximately a half

Step Chart

Sex	Seatbelt
Male	Rarely
Female	Mosttimes
Female	Always
Male	Always
Female	Always
Female	Always
Female	Sometimes
Female	Always
Female	Mosttimes
Female	Rarely
Female	Mosttimes
Female	Always
Female	Mosttimes
Male	Always
Male	Mosttimes

□ What is the Mode for Seatbelt variable?

- Recall: the most often seen label
- Where do we get this information? Answer: the frequency table

Seatbelt	Frequency
Never	0
Rarely	2
Sometimes	1
Mosttimes	5
Always	7

So, the mode for Seatbelt is:

Always

Mode of ordinal data

Sex	Seatbelt
Male	Rarely
Female	Mosttimes
Female	Always
Male	Always
Female	Always
Female	Always
Female	Sometimes
Female	Always
Female	Mosttimes
Female	Rarely
Female	Mosttimes
Female	Always
Female	Mosttimes
Male	Always
Male	Mosttimes

- What is the Median for Seatbelt variable?
 - Unlike nominal, ordinal data can be ordered!
 - Hence, we can compute the Median!
- How to compute median?
 - Arrange data in increasing order
 - Then, take the middle value!

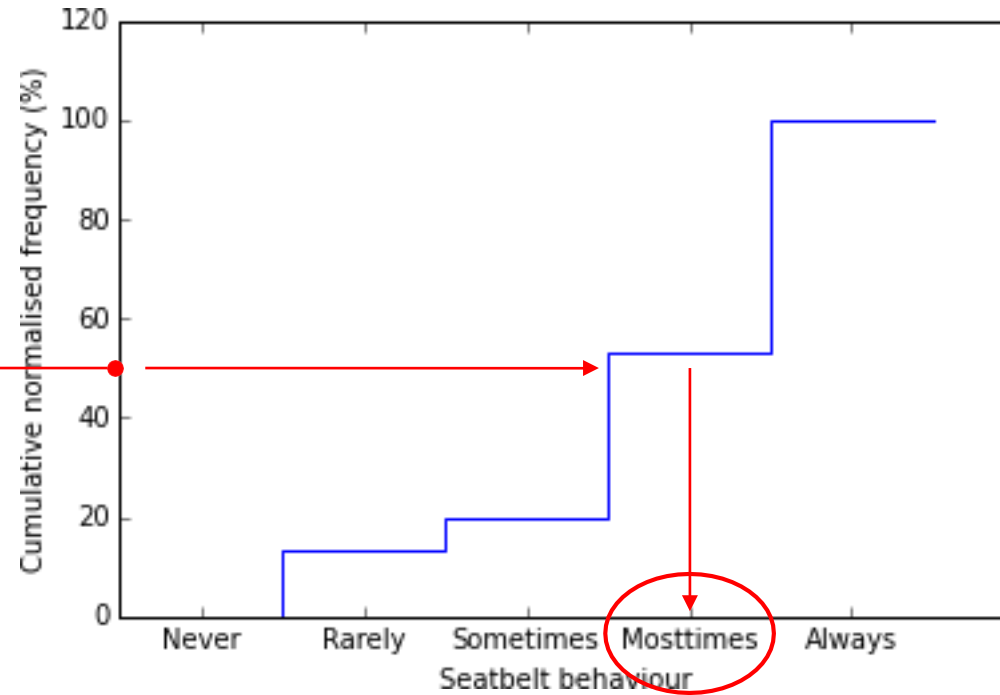
Seatbelt	Order
Never	1
Rarely	2
Sometimes	3
Mosttimes	4
Always	5

Median of ordinal data

Sex	Seatbelt
Male	Rarely
Female	Mosttimes
Female	Always
Male	Always
Female	Always
Female	Always
Female	Sometimes
Female	Always
Female	Mosttimes
Female	Rarely
Female	Mosttimes
Female	Always
Female	Mosttimes
Male	Always
Male	Mosttimes

- Isthere another way to determine the median?
- Yes: use the step chart for cumulative normalised frequency.

Determine the 50% point



Median of ordinal data

Summary for Ordinal Data

□ Numeric description:

- Frequency
- Normalised Frequency
- Cumulative Frequency
- Cumulative Normalised Frequency.
- Mode, ~~Mean~~, Median

□ Visual description:

- Pie Chart
- Tag Cloud
- Bar Chart
- Clustered/Stacked Bar Chart
- Step Chart

Understand Discrete Data

□ Numeric:

- Frequency
- Normalised Frequency
- Cumulative Frequency
- Cumulative Normalised Frequency.
- Mode, **Mean**, Median

□ Visualization:

- Pie chart(rarely)
- Bar Chart
- Tag Cloud(rarely)
- Clustered/Stacked Bar Chart.

	A	B	C	D	E
1	ParticipantID	Gender	Age	Hair Color	Height (m)
2	1	Male	18	Black	1.8
3	2	Female	14	Red	1.57
4	3	Male	17	Black	1.73
5	4	Male	22	Black	1.7
6	5	Male	23	Black	1.81
7	6	Female	21	Brown	1.63
8	7	Male	23	Red	1.7
9	8	Female	21	Brown	1.55
10	9	Female	19	Red	1.62
11	10	Female	20	Black	1.6
12	11	Male	18	Brown	1.71
13	12	Male	25	Brown	1.65
14	13	Female	13	Brown	1.56

- What is the Mean for Age variable?
 - Simply the average!
 - It is the most frequently descriptive statistics for metric data.
 - Why it doesn't make sense to compute the mean for categorical data?
 - Because, by definition, they cannot be placed on a number line!

$$\frac{18 + 14 + 17 + 22 + 23 + 21 + 23 + 21 + 19 + 20 + 18 + 25 + 13}{13} = 18$$

— Average age of the participants

The Mean for metric data

Understand Continuous Data

□ Numeric:

□ Grouped Representation

- Frequency
- Normalised Frequency
- Cumulative Frequency
- Cumulative Normalised Frequency.

□ ~~Mode~~, Mean, Median

□ Five-number summary.

□ Visualization:

□ Box-Plot

□ Histogram = bar chart

	A	B	C	D
1	Participant	Gender	Hair Color	Height (m)
2	1	Male	Black	1.80
3	2	Female	Red	1.57
4	3	Male	Black	1.73
5	4	Male	Black	1.70
6	5	Male	Black	1.81
7	6	Female	Brown	1.63
8	7	Male	Red	1.70
9	8	Female	Brown	1.55
10	9	Female	Red	1.62
11	10	Female	Black	1.60
12	11	Male	Brown	1.71
13	12	Male	Brown	1.65
14	13	Female	Brown	1.56

□ Grouped Representation

- Divide data into consecutive intervals of values
- Turn continuous data to ordinal data, hence we can apply numeric and visual description!

Height	Frequency	Normalised Frequency(%)	Cumulative Frequency	Cumulative Normalised Frequency(%)
1.4 – 1.6	3	23.1	3	23.1
1.6 – 1.7	6	46.1	9	69.2
1.7 – 1.8	3	23.1	12	92.3
1.8 – 1.9	1	7.7	13	100
> 1.9	0	0	13	100

Grouped Representation

	A	B	C	D
1	Participant	Gender	Hair Color	Height (m)
2	1	Male	Black	1.80
3	2	Female	Red	1.57
4	3	Male	Black	1.73
5	4	Male	Black	1.70
6	5	Male	Black	1.81
7	6	Female	Brown	1.63
8	7	Male	Red	1.70
9	8	Female	Brown	1.55
10	9	Female	Red	1.62
11	10	Female	Black	1.60
12	11	Male	Brown	1.71
13	12	Male	Brown	1.65
14	13	Female	Brown	1.56

Five-number summary

- Minimum: the smallest value
- Lower Quartile: (25%)
- Median: (50% or middle value)
- Upper Quartile: (75%)
- Maximum: the largest value

Much better to use computer rather than manually done by hand!

```
import pandas as pd

height = pd.Series([1.8, 1.57, 1.73, 1.7, 1.81,
                    1.63, 1.7, 1.55, 1.62, 1.6,
                    1.71, 1.65, 1.56])

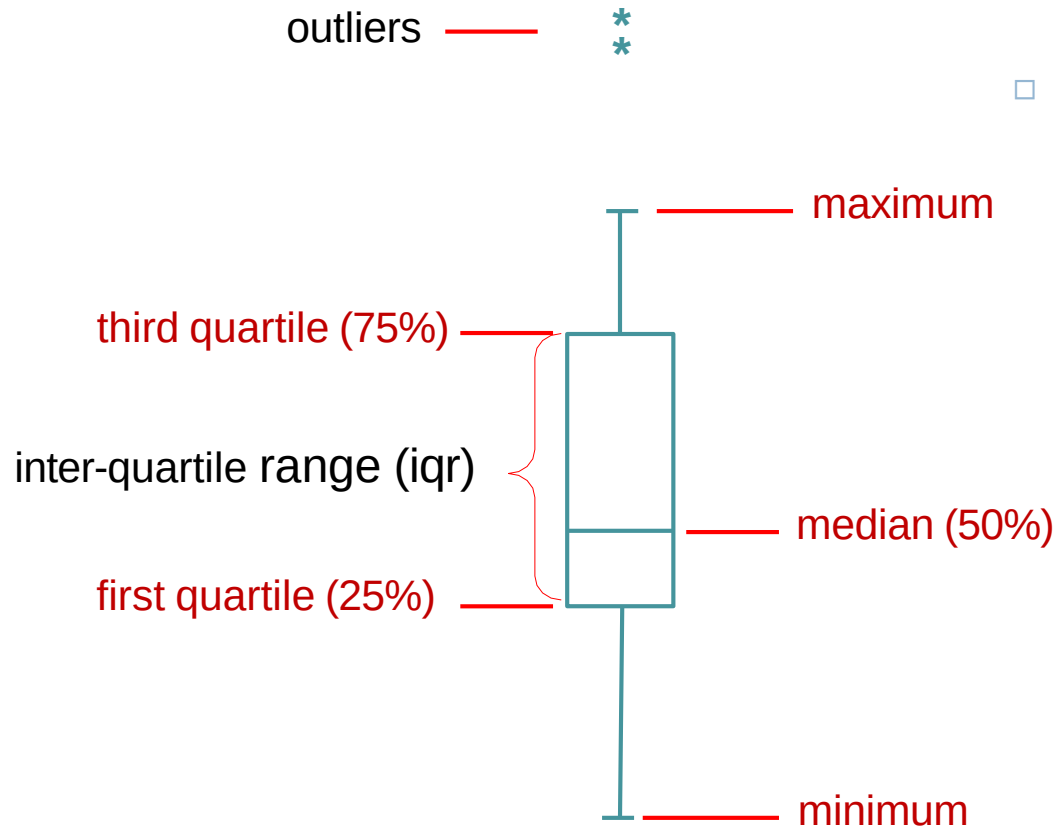
height.describe()
```

```
count    13.000000
mean     1.663846
std      0.086075
min      1.550000
25%      1.600000
50%      1.650000
75%      1.710000
max      1.810000
```

Five-number summary

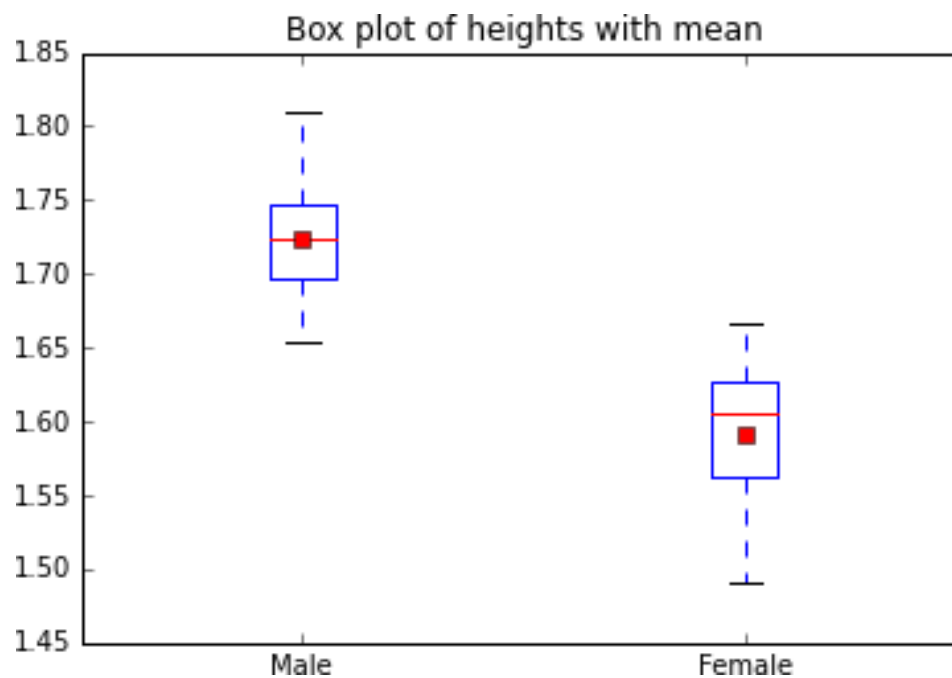
□ Boxplot

- Similar to a visual display of five-number summary without the mean.
- Extra information can be added, such as outliers.



Boxplot

FILE HOME INSERT PAGE LAYOUT FORMULAS DATA				
F10				
	A	B	C	D
1	Participant	Gender	Hair Color	Height (m)
2	1	Male	Black	1.80
3	2	Female	Red	1.57
4	3	Male	Black	1.73
5	4	Male	Black	1.70
6	5	Male	Black	1.81
7	6	Female	Brown	1.63
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10	9	Female	Red	1.62
11	10	Female	Black	1.60
12	11	Male	Brown	1.71
13	12	Male	Brown	1.65
14	13	Female	Brown	1.56

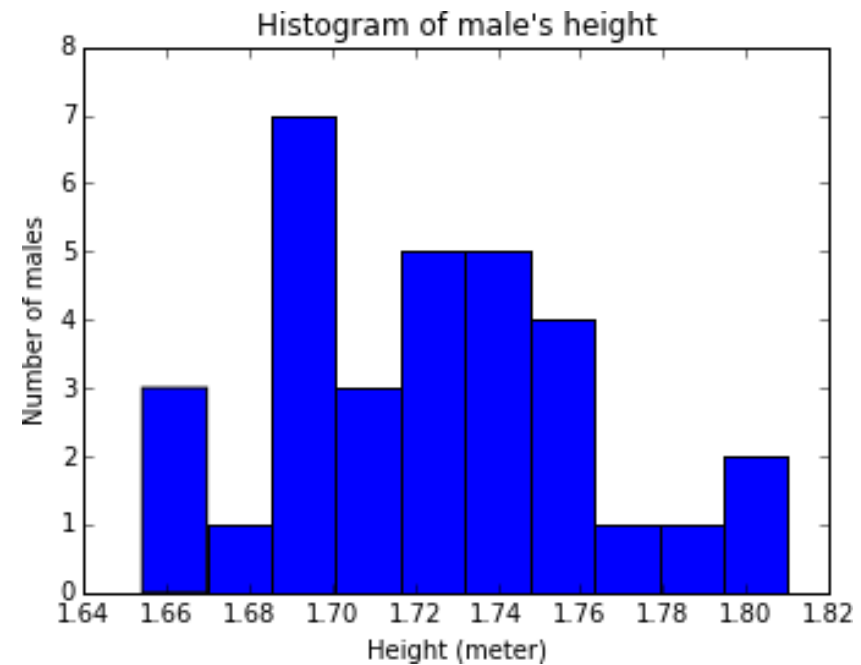


Box-Plot

Participant	Gender	Hair Color	Height
1	Male	Black	1.797692
2	Female	Red	1.567089
3	Male	Black	1.734498
4	Male	Black	1.696538
5	Male	Black	1.810248
⋮	⋮	⋮	⋮
46	Male	Red	1.662167
47	Male	Brown	1.696404
48	Female	Brown	1.597542
49	Male	Black	1.723152
50	Female	Black	1.609744

□ Histogram

1. Divide data into consecutive bins of equal size.
2. Count the number of data points in each bin.
3. Plot a bar chart from the resulting data.

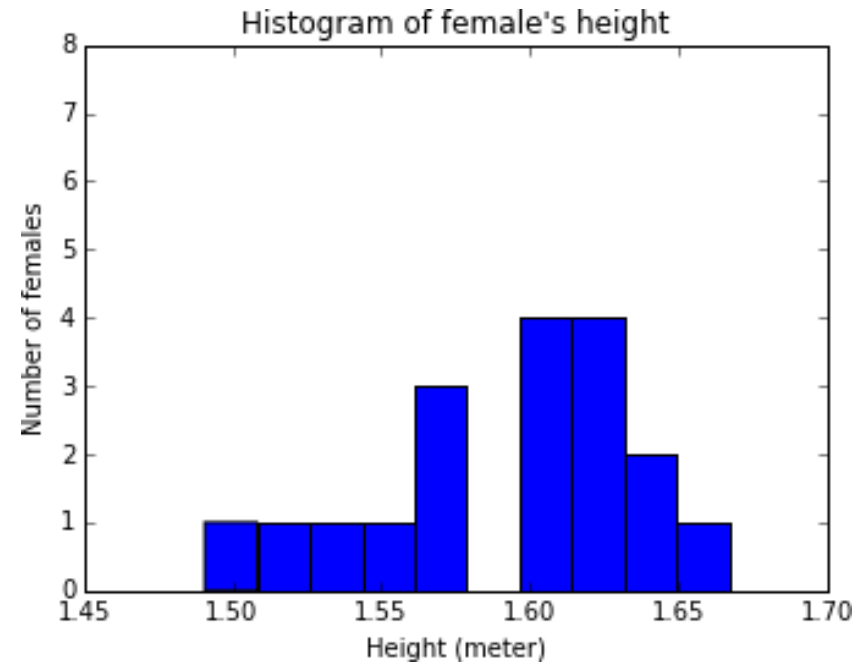


Histogram

Participant	Gender	Hair Color	Height
1	Male	Black	1.797692
2	Female	Red	1.567089
3	Male	Black	1.734498
4	Male	Black	1.696538
5	Male	Black	1.810248
⋮	⋮	⋮	⋮
46	Male	Red	1.662167
47	Male	Brown	1.696404
48	Female	Brown	1.597542
49	Male	Black	1.723152
50	Female	Black	1.609744

□ Histogram

1. Divide data into consecutive bins of equal size.
2. Count the number of data points in each bin.
3. Plot a bar chart from the resulting data.

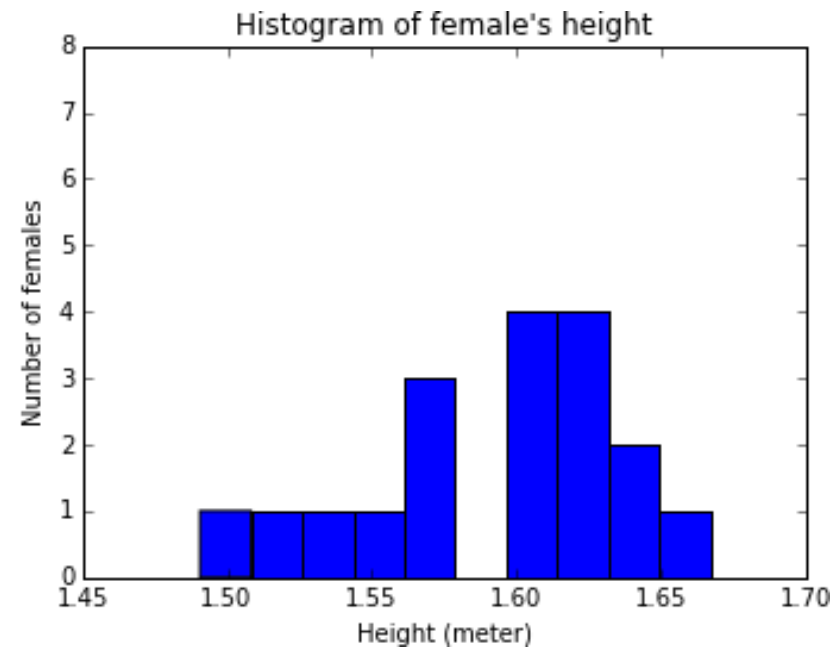
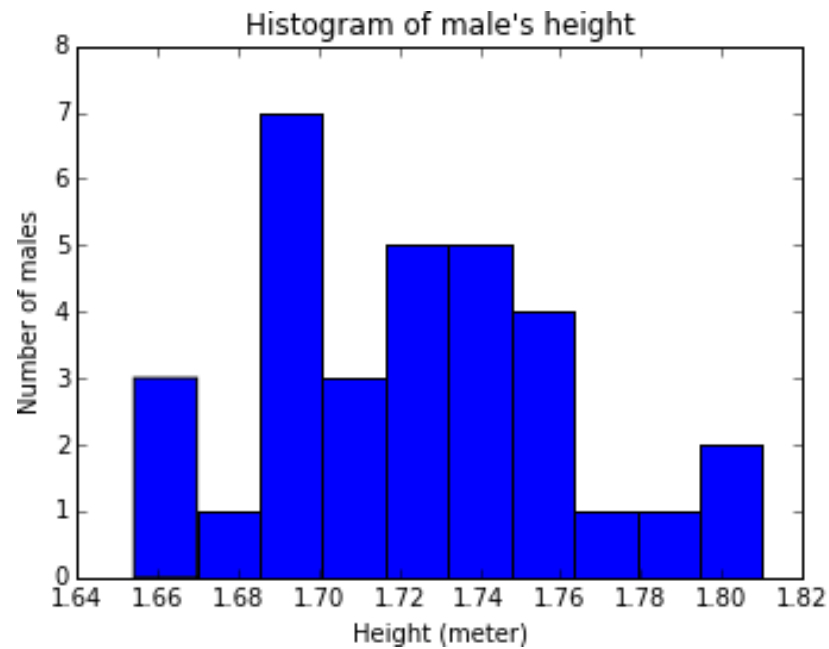


Histogram

□ Histogram

- Put them side by side for comparison.

1. Divide data into consecutive bins of equal size.
2. Count the number of data points in each bin.
3. Plot a bar chart from the resulting data.

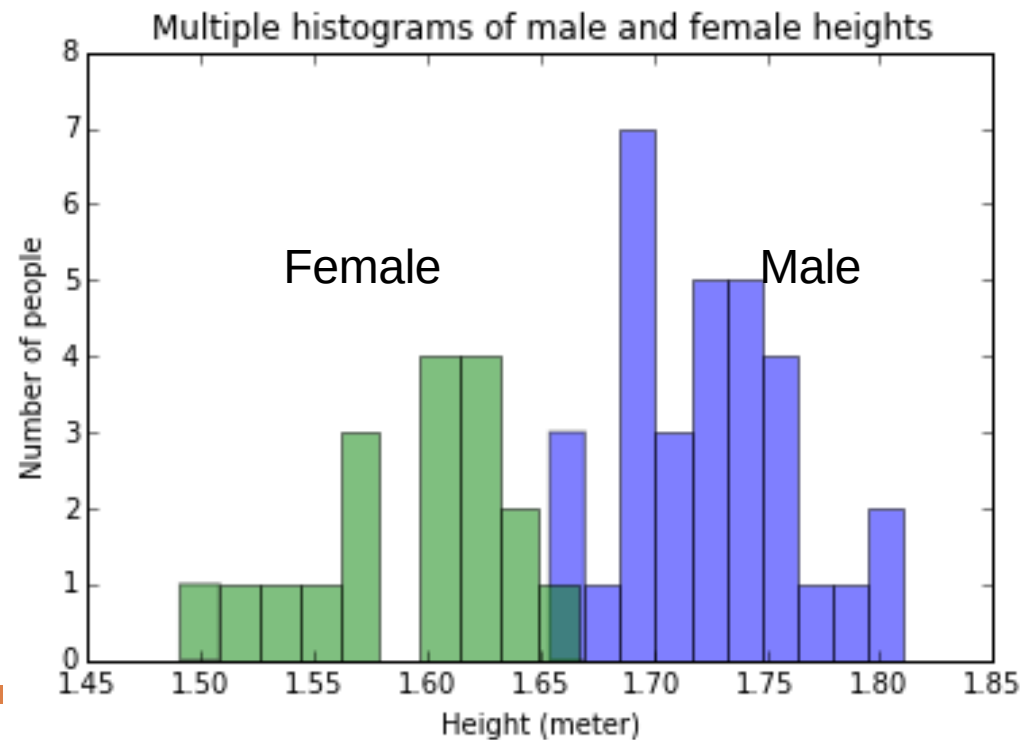


Histogram

□ Histogram

- Or, we put plot them together.
- We call this multiple histograms

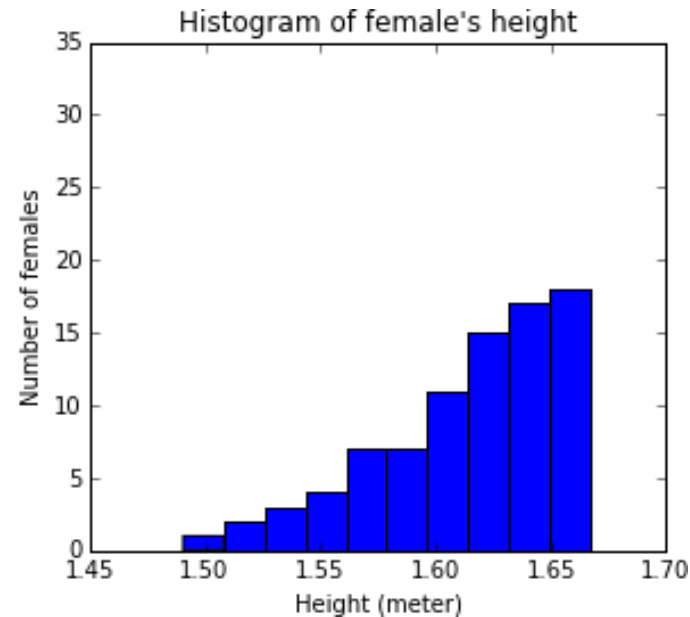
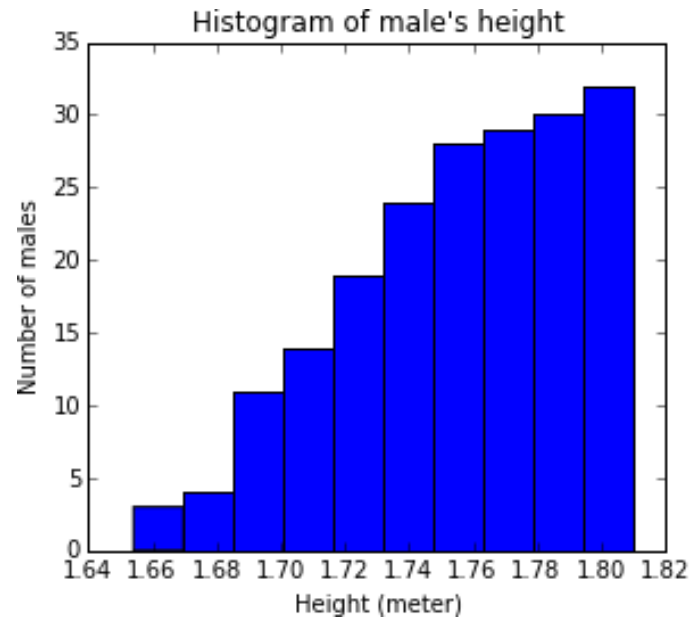
1. Divide data into consecutive bins of equal size.
2. Count the number of data points in each bin.
3. Plot a bar chart from the resulting data.



Histogram

□ Cumulative Histogram

- Cumulative plots are also applicable:
 - cumulative histogram
 - cumulative normalised histogram
 - cumulative distribution (= step chart for categorical data)



Cumulative Histogram

Summary for Continuous Data

□ Numeric description:

□ Grouped Representation

- Frequency
- Normalised Frequency
- Cumulative Frequency
- Cumulative Normalised Frequency.

□ ~~Mode~~ (but, possible with proper distribution), Mean, Median

□ **Five-number summary.**

□ Visual description:

□ **Box-Plot**

□ Histogram = bar chart

Which chart to choose?

Attribute Type		Pie Chart	Tag Cloud	Bar Chart	Histogram (grouped)	Step Chart	Cumulative
Categorical	Nominal	yes	yes	yes	no	no	no
	Ordinal	possible	yes	yes	no	yes	possible
Metric	Discrete	no	possible	possible	yes	yes	yes
	Continuous	no	no	No, unless grouped	yes	no	yes

possible* : it is possible to draw the chart, but not recommended in general case.

Reference

- *NIST Big Data Interoperability Framework, Vol 1, Definitions.*[**NBDRA-1**]
<http://dx.doi.org/10.6028/NIST.SP.1500-1>
- *NIST Big Data Interoperability Framework, Vol 2, Big Data Taxonomies* [**NBDRA-2**]
<http://dx.doi.org/10.6028/NIST.SP.1500-2>
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